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FUNDAMENTALS OF WHOLE BRAIN EMULATION: STATE, TRANSITION AND UPDATE REPRESENTATIONS

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Whole brain emulation aims to re-implement functions of a mind in another computational substrate with the precision needed to predict the natural development of active states in as much as the influence of random processes allows. Furthermore, brain emulation does not present a possible model of a function, but rather presents the actual implementation of that function, based on the details of the circuitry of a specific brain. We introduce a notation for the representations of mind state, mind transition functions and transition update functions, for which elements and their relations must be quantified in accordance with measurements in the biological substrate. To discover the limits of significance in terms of the temporal and spatial resolution of measurements, we point out the importance of brain region and task specific constraints, as well as the importance of *in-vivo* measurements. We summarize further problems that need to be addressed.

Keywords: Whole brain emulation; functions of mind; resolution; empirical data; *in vivo* measurement.

1. Aiming for a Brain Emulator

The concept of brain emulation is essentially the same as that of well-known emulators for computer hardware. An emulator replicates the functions of the emulated system by using the computational hardware of another system, and it strives to do this so well that the emulated system behavior is indistinguishable from its behavior on original hardware. With this aim, we therefore implicitly presuppose that brain behavior is computable. In practice, most researchers interested in brain emulation assume that brain behavior can be expressed in the form of effectively calculable functions for which the Church-Turing thesis applies[1] [Church, 1932, 1936a,b, 1941; Turing, 1937a,b]. As in the case of other emulators, the highest priority is to insure that the results of processing on the emulator are the same as processing on the original hardware. While brain emulation provides access to the operations taking

[1]This in no way excludes emulation with the aid of quantum computers, as the Quantum Strong Church–Turing thesis [Kaye *et al.*, 2007] subsumes the Church–Turing thesis.

place during mental processes, that does not immediately give us an understanding of the strategies used at different levels of abstraction of functions of mind.

In the case of a human brain, we will use "mind" to refer to those functions that determine behavioral responses of the system. This involves processing environmental stimuli, stimuli elicited by intrinsic drives, spontaneous thought processes elicited by evoked memories, etc. Whole brain emulation means that the functions of mind are implemented in a substrate other than the original biological substrate. Brain emulation strives to achieve a mechanistic re-implementation, which makes it possible to predict an active state and behavior at time $t + \Delta t$ with acceptable error, if we know the state at time t. Our understanding that it is possible to re-implement functions of mind in another substrate [Levin, 2010] implies a form of functionalism that includes a functionalist interpretation of consciousness. Brain emulation is a young and in some circles still contentious aim, so that it is worthwhile to spend a considerable part of this paper presenting the fundamental practical issues of the topic.

The functionalist understanding of the mind is essential for emulation approaches. Furthermore, we must posit that there is a resolution of the mechanics of computation in the brain at which we can separate the functions being carried out from the physical materials that used to implement the computational operations. Were this not the case, then a precise emulation through any re-implementation other than an exact (sub-)atomic replica of the biological constituents would be impossible [Koene, 2011]. Consider a situation in which the absence of a robust, error correcting computational layer meant that any most minute difference of the effects that the computational substrate has on its environment were immediately catastrophic to mind or consciousness. Carrying out computations in different substrates is then equivalent to coding samples by using binary codes from different probability distributions. Expressed as a coding problem, the Kullback−Leibler divergence demonstrates the information theoretic overhead for that extreme case of emulation [Kullback and Leibler, 1951].

Fortunately, the fragile circumstance described in the preceding paragraph is unrealistically extreme. When we discuss emulators of computer hardware that is obvious: We care only if a program run on the emulated computer produces the same relevant results as it does when run on the original computer hardware. We are generally not interested in the totality of effects, such as the manner in which the original hardware distributes electrical current or dissipates heat. Similarly, a brain emulation should not strive to defeat its purpose by copying even undesirable dependencies of the original implementation, which led to the difficulty of accessing its operations without disturbing them in the first place.

For computer scientists and artificial intelligence researchers, an important reason to consider brain emulation is the opportunity to apply and to learn about an intelligent and generalistic system that is known to work. Brain emulation can supply proper grounding for many assumptions and hypotheses in neuroscience and artificial intelligence research. The resulting system will allow access to its internal operations

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to a degree that is extremely difficult to achieve in the biological substrate. In the clinical realm, brain emulation in whole or part are approaches to the development of neuroprostheses. The emphasis is then clearly on restoration of function, with ample resemblance to the computer emulation analogy above.

Here we focus on the objective of *whole* brain emulation, the emulation of all the relevant systems that support functions of the mind. We do this to take into account the modular yet interdependent nature of all functions that underlie behavior. Human brain architecture is inherently modular, but internal awareness and behavioral responses are the result of collaboration between many of those modules. Within each module, the functions depend on dynamic patterns of activity within large populations of neurons, and despite this, some functions, such as certain perceptual capabilities, depend crucially on details of the responses of individual neurons (e.g., auditory processing in the barn owl described by Gerstner *et al.* [1998]). Investigating functions of the mind therefore poses a multi-scale problem. We discuss human whole brain emulation, since initial studies aimed at small parts of the brain or at the smaller brains of other animals may offer similar neurophysiological challenges without constituting significantly different conceptual problems.

Furthermore, we focus on subject specific brain emulation. That constraint makes the project of whole brain emulation significantly different than large-scale brain simulation projects such as the Blue Brain Project [Markram, 2006]. The data required is also significantly different than the data collected in brain maps such as the CoCoMac database [Kötter, 2004] or the Allen Mouse Atlas [Lein *et al.*, 2006]. For a credible re-implementation of functions of mind, we must focus on the emulation of circuit structure and parameter values extracted from a specific brain. The combined structure and function information acquired from a specific brain is intrinsically meaningful in the sense that it relates directly to operational circuitry producing characteristic responses influenced by subject specific prior experience. We know little about the generally shared versus individually distinct features of brain anatomy, brain physiology or brain dynamics. Until we know more about the levels at which data is significant for function, subject specific analysis will be a prerequisite for the re-implementation of sensible neuronal circuit function. A more detailed discussion of the relevance of this difference between brain emulation and stochastic methods for the generation of species-specific brain models appears in Koene [2012a,c]. Subject specific brain emulation is directly applicable to function restoration through neuroprosthesis.

2. Data for a Brain Emulation

In order to discuss the data requirements for brain emulation we need a terminology with which to describe the state and evolution of functions of the mind. In computational neuroscience, it is common to use representations based on Hodgkin–Huxley channel equations [Hodgkin and Huxley, 1952], calcium dynamics (e.g., Caesar *et al.* [1993] or Harris [1999]) and exponential functions for the current or conductance

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response to synaptic input. Here we prefer a more general system representation, since we have yet to establish which biological components we are interested in and what the relevant biophysical functions are.

In computer science and artificial intelligence, it is popular to use information theoretic representations [Yu *et al.*, 2010] in terms of Markov processes. Here we can consider continuous-time Markov processes and state space. If we assume that the mechanistic composition of the brain mandates a finite state space, then we can also assume a transition matrix with which to represent the transition probability distribution. Changing transition matrices can take into account processes such as learning and memory. Similar terminology appears in computational neuroscience, for example in the case of the spike timing dependent neural coding work by Bialek and Rieke [Bialek *et al.*, 1991; Bialek, 1992; Rieke *et al.*, 1997], in which the likelihood of different spike trains t_i given stimulus $s(\tau)$ is described by moments of a probability distribution $P[t_i|s(\tau)]$.

Alternatively, applying the terminology of optimal control theory to our physical system [Eliasmith, 2008] we can posit a Hamiltonian $H(x, \lambda, u, t) = \lambda^T(t) f(x, u, t) + L(x, u, t)$, with state, $x(t)$, and costate variables, $\lambda(t)$, being vectors of equal dimension.[2] If we assume that there is a representation of mind in terms of a state vector $\mathbf{x}(t)$, then we can write a straightforward nonlinear state-space model,

$$\dot{\mathbf{x}} = \mathbf{f}(t, x(t), u(t)), \tag{1}$$

$$\mathbf{y} = \mathbf{h}(t, x(t), u(t)), \tag{2}$$

with system input vector $\mathbf{u}(t)$ and system output vector $\mathbf{y}(t)$. The system is not time-invariant, since changes in the active brain (e.g., modified neuronal cell morphology, modified chemical densities) imply changes of the state transition matrix.

When we seek to create an emulation of a (whole) brain, what we want is to retain the functioning mind, separated from its original dependence on a specific underlying biological substrate. Using the control theory inspired terminology established above, we distinguish the following data:

(1) **Mind State**, $\mathbf{x}(t)$: Qualitatively, this is a snapshot of the mental activity that elicits internal awareness and external behavior (a generalization of the arguments in Brown [2006]). We presuppose that both internal awareness and external behavior are biophysically mediated by electrical and chemical signals.

(2) **Mind Transition Functions**, $\mathbf{f}(t, x(t), u(t))$: These are functions that describe the dynamic development of the Mind State. With these transition functions we should be able to predict (with acceptable deviation, $\epsilon < \epsilon_{\max}$, and for a restricted time interval, $\Delta t < \Delta t_{\max}$) what a Mind State $\mathbf{x}(t + \Delta t)$ should be, given a known Mind State $\mathbf{x}(t)$. The matter of selecting among possible

[2] Optimal control theory provides a useful and succinct notation applied to stochastic brain mechanisms, even without presupposing control of actions by near-optimal brain behavior, as in Doya *et al.* [2007].

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representations for an emulation and the relationship of that choice with regards to deviation over a limited time interval is addressed in Koene [2012b].

(3) **Transition Update Functions**, $\mathbf{g}(t, f(t), x(t), u(t))$: These describe how Mind Transition Functions change, for example due to modifications caused by mental activity. Examples of change in the biological substrate are modified neurotransmitter vesicle concentrations, modified local calcium concentrations, modified axo-dendritic structure, synaptogenesis, neurogenesis, etc. This includes all forms of plasticity. The changes appear in the functions $\mathbf{f}(t, x(t), u(t))$, and they constitute a memory or history of Mind State.

To acquire (also popularly expressed as "to upload" or "to transfer") the functions of mind and to enable an emulated implementation thereof consists of two crucial efforts:

(1) Identify the sets of elements and relations that are expressed in $\mathbf{x}(t)$, $\mathbf{f}(t, x(t), u(t))$ and $\mathbf{g}(t, f(t), x(t), u(t))$.
(2) Quantify and measure the parameter values of those state, transition and update representations.

Data acquisition can involve different degrees of precision, different spatial and temporal resolutions. Optimally, we wish to acquire the minimum amount of data from the brain that enables us to retain the functioning mind in information theoretic terms. Notice that this is a different problem than questions such as *what is the minimum information that needs to be stored* or *what is the most efficient process required to represent functions of the mind*. After all, the biological implementation in which we search for parameter values is a given.

Consider the question of the resolution at which sufficient data is made apparent to account for the information mediated by electrical and chemical signals, modified according to transition and update functions: Localized brain activity (e.g., voxels), activity of groups of neurons, time-averaged (rate) activity of individual neurons, spiking activity of individual neurons, (analog) membrane response to activity in individual neurons, (spatially specific) responses throughout and around neuronal morphology, or responses at the molecular level. These are all significant concerns to which we can devote many future papers.

Here, we note that at each candidate resolution information is contained both in the characteristic parameters of dynamic physiological functions and in the structure that governs interactions between components. (This is the more general case of the well-known function-structure entanglement in neuronal networks.) We need to understand the significance of both structure and function data at different resolutions.

For practical purposes, it is useful that high resolution knowledge about function can enable us to infer knowledge about structure and vice versa. For example, if we understand detailed transition functions at the level of individual neurons ($\mathbf{f}_\nu(\cdot)$) and

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we also obtain the input−output transition matrix derived from recorded stimulus responses of the network of neurons ($\mathbf{u}_\Psi(t)$ and $\mathbf{y}_\Psi(t)$) then we can deduce the connectivity between the individual neurons that explains the observed stimulus-response set [Rigat *et al.*, 2006].

3. Discovering the Resolution

Carrying on with the structure-function duality, we propose simplifications of the problem. Consider the elements and relations of $\mathbf{x}(t)$, $\mathbf{f}(t, x(t), u(t))$ and $\mathbf{g}(t, f(t), x(t), u(t))$ separately for:

(1) Different functional regions of the brain.
(2) Different mental functions or tasks.

Functional brain regions are subject to different constraints in terms of the input they receive and the output they deliver [Zhou *et al.*, 2007]. Additionally, the biophysical system within a brain region is constrained within limits of computational precision (e.g., due to noise, response timing, required tolerance to parameter variation, requirements for robust performance), exhibiting intermittent stochastic, periodic and chaotic modes of operation [Freeman and Yao, 1990; Freeman, 1991; Moreira and Andrade, 1994; van Vreeswijk and Sompolinsky, 1996]. Knowing these constraints and the consequent maximum required or achievable densities of information processing can help identify limits for the significant resolution of elements.

Similarly, different tasks require different information processing, in terms of quantity, precision and rate. Knowledge of these requirements can provide lower and upper bounds, for example, for the spatial and temporal resolution that are required during functional recording[3] in order to quantify $\mathbf{x}(t)$, $\mathbf{f}(t, x(t), u(t))$ and $\mathbf{g}(t, f(t), x(t), u(t))$. The discovery of operational constraints is a prime candidate for modeling studies [Bialek *et al.*, 1991], but depends critically on experimental measurements that are made *in-vivo*.

3.1. *In vivo versus post-mortem*

A number of projects are ongoing that directly contribute to the two crucial efforts enumerated above. Part (a), the identification of elements and relations that are relevant to functions of mind, is addressed by projects that test hypotheses about the descriptive resolution that is needed, and by projects that test our ability to infer correct combinations of functions from the data that we are able to acquire at large scale and high resolution. In the first category, we have projects such as the work by David Dalrymple and colleagues at Harvard and MIT to functionally characterize

[3] The resolution requirements for $\mathbf{x}(t)$, $\mathbf{f}(t, x(t), u(t))$ and $\mathbf{g}(t, f(t), x(t), u(t))$ need not be identical. In fact, an early poll of expert opinions points to a general, but speculative, consensus that quantification of $\mathbf{f}(t, x(t), u(t))$ demands higher resolution data than quantification of $\mathbf{x}(t)$.

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all 302 neurons of the nematode. Elegans in order to produce a verifiable whole brain emulation of that animal. In the second category, we find recent result such as those by Briggman *et al.*, in which large-scale functional characterization of the retina was paired with the high-resolution acquisition of the structural connectome. Registration of the structural and functional maps showed that it was possible to deduce function from structure within the context of the experiment [Briggman *et al.*, 2011]. Beyond individual experiments, there are also systematic efforts to further explore the functional involvement and relevance of biological constituents within the connected structure of the brain, such as the Allen Brain Atlas and new large-scale high-resolution exploration of the mouse visual system at the Allen Brain Institute [Jones *et al.*, 2009].

Part (b), developing methods for the quantification and measurements of parameter values at the levels of description that have been identified as relevant, is addressed by projects that are developing next-generation tools for neuroscience. Advanced *post-mortem* techniques are being developed with which to extract structural data at large scale and high resolution, for example using the Knife-Edge Scanning Microscope (KESM) technology [McCormick and Mayerich, 2004] or the Automated Tape-Collecting Lathe Ultra-Microtome (ATLUM) technology [Hayworth *et al.*, 2007]. Both are designed to section a whole brain at a resolution that enables light or electron microscope imaging to collect image stacks for the whole brain. The structure of the neuronal network of that specific brain, i.e., the so-called connectome, can be reconstructed from that data. If one relies solely on structure data, then all functional data must be inferred. The quality and applicability of the inferences directly determines the performance of a resulting emulation.

In practice, this involves analysis of three-dimensional reconstructions of the morphology of those components of the biological specimen that were visible to the scanning procedure used, making decisions based on correlation with the morphology of known components (e.g., neuron types), in order to select state, transition and update functions to use in a functional reconstruction. In the case of KESM (Fig. 1), the visible subset also depends on contrast dyes used and on the distribution of those dyes.

We may not need to rely solely on structure data, as methods to obtain whole brain read-out of dynamic activity are also emerging. The synthetic neurobiology group at MIT, led by Ed Boyden, aims to leverage its optogenetic tool developments into the domain of large-scale dynamic recording *in vivo*. That is possible by pairing optogenetic stimulation with large arrays of recording electrodes. Large bandwidth recording from many neurons is also being addressed through the development of new voltage sensitive proteins in ongoing efforts of Adam Cohen's group at Harvard. Beyond this, early collaborative research involving groups such as Konrad Kording's at Northwestern University are designing molecular methods for the intra-cellular recording of dynamic activity at large scale and high resolution.

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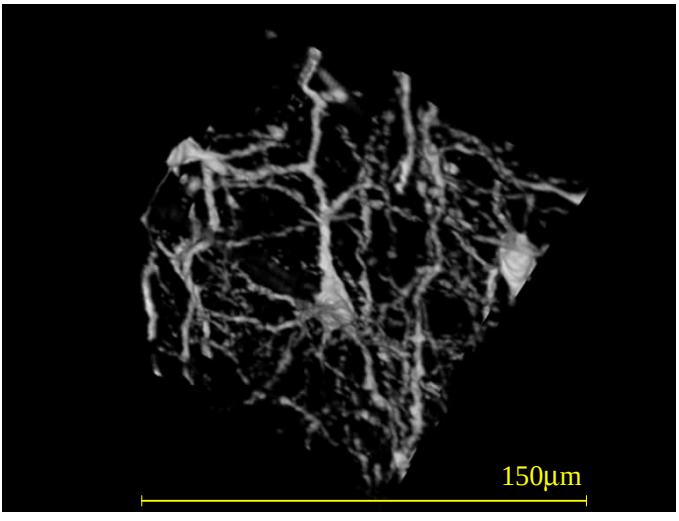


Fig. 1. Excerpt from an animation that demonstrates the three-dimensional reconstruction from images taken by the Knife-Edge Scanning Microscope. The sample shown contains the partial reconstructed morphology of cortical pyramidal cells, from a whole mouse brain Golgi-stained data set. (Courtesy of Yoonsuck Choe, Brain Networks Laboratory, Texas A&M University.)

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3.2. *Combining large scale and high resolution*

Ideally, we would use in a brain model only parameter values that we can measure directly. We would like to carry out actual measurements of structural morphology and actual recordings of functional activity. Consider, for example, the inference of functional connectivity that can be made by observing currents in several components simultaneously. The result can easily miss latent function if the observations are carried out over a small finite time span. Actual structural measurements can improve the connectivity data. Similarly, functional data can help to correct assumptions about the type and parameter values of a component that were made by inferring from component morphology. In a highly over-parametrized system, such as an individual brain, these are ways to constrain the prediction error of our emulation in the context of responses to $\mathbf{x}(t)$ and $\mathbf{u}(t)$ never observed in the recorded data set.[4] Quantifying which deviations of predicted response from the actual response that the biological implementation would produce are acceptable is a topic worthy of attention in future papers.

Low-resolution, large-scale data: Whole-brain data is presently obtainable not only through *post-mortem* techniques, but also through functional scans with comparatively low resolution, e.g., (functional) magnetic resonance imaging (f)MRI. Recordings made during these scans inspect physiological manifestations of a subset

[4]See for a comparison efforts to achieve spike train prediction through single neuron modeling [Gerstner and Naud, 2009].

of activity that is visible in the scanning spectrum used (e.g., blood-oxygen-levels in fMRI). Due to low temporal as well as spatial resolution, scans generally identify low frequency components of the visible dynamics. It is possible to infer some of the functional connectivity, the human connectome [Sporns *et al.*, 2005, 2007], based on observed large-scale activity, using electroencephalography (EEG), magnetoencephalography (MEG), fMRI and advances in diffusion-tensor MR imaging techniques (DTI), as exemplified by Fig. 2. DTI can provide information about bundles of axon fiber, the brain's white matter. It is important to realize that in addition to the low resolution of data acquired, and visibility problems (e.g., seeing only differences in blood-oxygen-levels), recording can usually take place only for very limited amounts of time. Experiments of limited duration expose only few of the responses that are supported by the true functional connectivity. Overall, it is clear that low-resolution, large-scale data acquisition of this sort enables only limited inference to the underlying complexity of the system.

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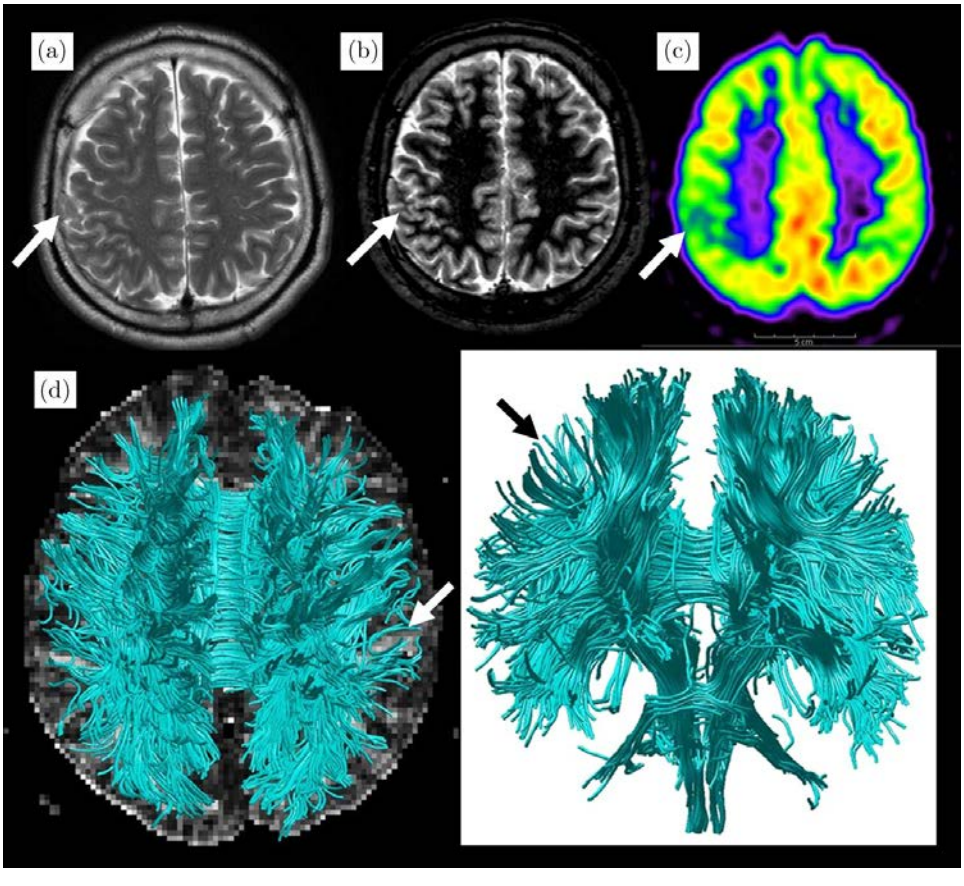


Fig. 2. Example of diffusion-tensor MR results exposing white matter tracts in the brain of a 17-year-old boy with left-side motor seizure, from Nucifora *et al.* [2007, Fig. 6].

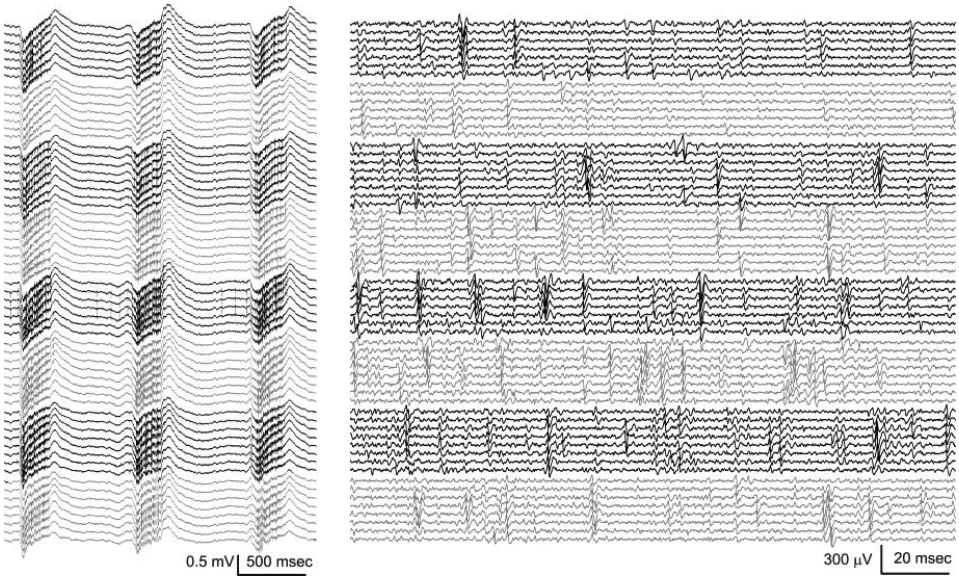


Fig. 3. Example of parallel recording of unit activity in layer V of rat somatosensory cortex with a 64 channel silicon probe, from Csicsvari *et al.* [2003, Fig. 3].

High-resolution, small-scale data: Sparse, but highly resolved functional data is obtained through implanted recording electrodes (or arrays of such electrodes), as demonstrated in Fig. 3 [Csicsvari *et al.*, 2003; Buzsáki, 2004]. Here also, the usefulness of the data acquired for inference with regards to connectivity is constrained by the duration over which and times at which recording experiments are carried out. Few interactions between system components can be identified in this manner. By analogy, analyzing electrode recordings carried out at several sites throughout the brain resembles an attempt to understand the content of a painting by a great master through the inspection of details in several arbitrarily located postage stamp sized excerpts. Some aspects of style and technique may be recognized, but Rembrandt's Night Watch remains unseen.

When we lack knowledge of structure this restricts our ability to predict possible patterns of activity. Consequently, there is limited extrapolation to unseen behaviors. If we can acquire data *in vivo* at large scale and high resolution, especially if the techniques used allow us to acquire both function and structure data, then the likelihood is greatly improved that we may successfully emulate system operations and predict responses for episodes not observed during experimental study.

When we can combine a full structure map of a connectome with a map of the characteristic functions of the components in that structure, then the prerequisites for whole brain emulation will have been achieved. The two critical directions of research are what we can distinguish as structual connectomics and functional

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connectomics. In both, the aims are brain-specific. Their combination is the acquisition of brain-specific circuitry.

For a useful, functioning mind operating in another substrate there are additional requirements, such as sensory input and output to effectors. We will also want to design computing platforms that more suitably fit the requirements of operational mind functions. Those topics deserve to be address, but they go beyond the scope of this paper.

3.3. *Measurement requirements*

We seek empirical grounding for structure and function parameters through direct measurements of morphology and functional correlates. Doing this reduces the number of free parameters to be inferred from recorded data. The characteristic requirements for these measurements, to be taken from a large number of widely distributed recording sites, are:

- *In vivo* recording of full episodes.
- *In vivo* measurement of physiological parameters.
- *In vivo* measurement of morphological parameters.

The current generation of *in vivo* interface techniques and recording devices cannot meet our requirements. Chronic electronic implants are still problematic, struggling with diminishing capabilities and necessitating follow-up operations. Recording is restricted to very small subsets of vast brain networks. Recording from the same cell is not sustained for long durations, and it is impossible to detect information that is not directly correlated with measurable electric currents. It is in part due to these limitations that present techniques are rarely used to analyze networks in systems that behave in real-world dynamic environments.

To answer element, relation and quantification questions identified above, the design of new *in vivo* measurement techniques used, if utilizing probes or imaging technology, will need to take into account the following performance requirements:

- Reliability of the interface, reliability of the scanning technology.
- Very large-scale neural recording or interfacing.
- Recording or interfacing sites distributed throughout the brain, brain-wide scans.
- High rate of data acquisition, in the order of the maximum frequency of the dynamics exhibited.
- High spatial or component resolution during data acquisition.

Important information about any measurement are time and place of the measurement. We may substitute registration of the relative locations of measurements with respect to each other for absolute location, and we may substitute the order of regular measurements for absolute time registration if necessary. We present several interesting avenues for the design of suitable techniques in a related paper [Koene, 2012b].

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4. Discussion

The fundamental problem at the core of the whole brain emulation effort is the question: What is the significant information about functions of the mind that is contained in their implementation in the biological substrate? In this paper, we formulated this problem in terms of the need to identify sets of elements and relations that are expressed in state, $\mathbf{x}(t)$; transition, $\mathbf{f}(t, x(t), u(t))$; and update functions, $\mathbf{g}(t, f(t), x(t), u(t))$; and in terms of the need to quantify and measure their parameter values. We should also consider if the answers to these questions generalize to all brains, or which types of subject specific differences there may be. It is likely that the answers are not the same for all modular parts or functions of the brain, so that we are faced with a problem of integrating multiple scales.

With the advent of organized efforts in neuroinformatics [Koslow and Huerta, 1996; Arbib and Grethe, 2001; Koslow *et al.*, 2005] a number of research projects venture into realms of scale and resolution that may lead to some of the knowledge required for whole brain emulation. The Blue Brain project [Markram, 2006, 2008], led by Henry Markram, aims to reverse engineer the human brain down to the molecular level. At present, the project has created a simulation of a rat neocortical column containing approximately 10,000 neurons. The constructs simulated are based on architectural and functional principles identified in a large number of studies with many rat brains.

Large-scale brain models by Eugene Izhikevich [Izhikevich and Edelman, 2008] involve microcircuitry composed of neurons that are much simplified compared with neurons in the Blue Brain project. Those models achieved simulations of up to 100 billion neurons with 1 quadrillion synapses (see Fig. 4), which exhibit α and γ rhythms, as well as moving clusters of activity (but with some limits to state storage). Multi-center projects, such as the Self-Constructing Computing Systems (SECO) project (http://www.seco-project.eu/) and its precursors,[5] seek to explain by reverse engineering developmental processes at the cellular level and self-generational rules, how the neocortex can self-organize into its morphological and functional complexity, as well as how the same principles can be applied in artificial circuitry. Early results of research into cognitive neuroprostheses, such as effort in the laboratory of Ted Berger to create electronic circuitry to replace subnetwork specific function of the hippocampal structure [Hsiao *et al.*, 2009], demonstrate the need for these multi-scale investigations.

A fundamental formulation of the problem of whole brain emulation, as attempted here, is only the first of many crucial issues that deserve to be addressed. An obvious next question is, what techniques can be used as methods of data collection, with which it would be possible to identify and quantify parameters? Those methods will need to meet not only a set of safety and performance requirements, but may also need to cope with issues such as the generation or delivery of power to

[5]SECO builds on projects such as CASPAN (http://neurodynamics.nl/), including the design of the NETMORPH framework [Koene *et al.*, 2009].

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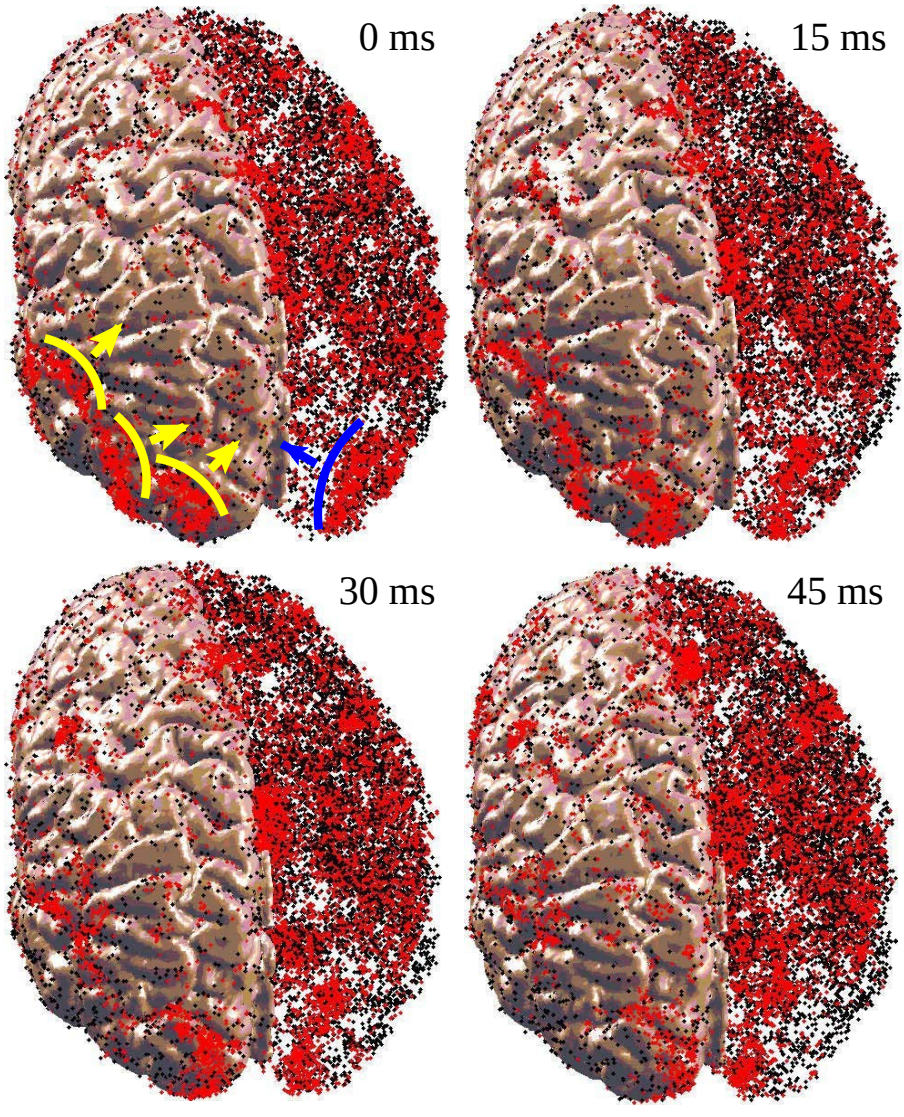


Fig. 4. Propagating waves of activity in Izhikevich's large-scale model of mammalian thalamocortical systems, from Izhikevich and Edelman [2008, Fig. 6].

measurement probes, size and complexity of the data stream, and dealing with adaptation of and changes in the biological medium during the period in which measurements are taken [Sporns *et al.*, 2005, p. 246].

When large amounts of empirical data are obtained, then we face the problem of properly analyzing all that data. Initially, we would like to use the data to establish operational constraints within parts of the biological substrate and within specific tasks. With such an analysis we can systematically establish relevant element sets

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and their relations for the functions $\mathbf{x}(t)$, $\mathbf{f}(t, x(t), u(t))$ and $\mathbf{g}(t, f(t), x(t), u(t))$. Relevant element sets will specify the type and resolution of components that contribute significant information to the system, as well as the significant resolution of morphological structure.

That is a large but finite task, which may be aided by new developments in data mining and data analysis (utilizing the power of current and emerging computing hardware, e.g., parallel processing architectures and quantum computers). An important part of this effort involves a determination of acceptable error: For a successful whole brain emulation, what is the acceptable deviation $\epsilon_{\max}$ of emulated state $\mathbf{x}(t + \Delta t)$ from the range of probable states that the original brain would have reached? The amplitude of the acceptable error affects the difficulty of data acquisition, as well as the objective quality of system emulation, particularly when the functions of mind of a specific subject are to be emulated. Ultimately, we need to be able to assess the precision of a brain emulation. Ideally, we would do this by looking at successive emulated brain states, though practically we may instead have to observe behavioral output that is correlated with the update of those states. The steps of a protocol are described in Koene [2012b].

Finally, future work can address the use of efficient, powerful or otherwise attractive new computational substrates for the implementation of emulated state, transition and update functions (e.g., in-silico Jackson *et al.* [2011]; Merolla *et al.* [2011]; Seo *et al.* [2011], photonic [Ibrahim *et al.*, 2004] or superconducting [Crotty *et al.*, 2010] neuromorphic circuit architectures), providing platforms for the possible enhancement and augmentation of functions of the mind. Implementation will depend on algorithms for large-scale reconstruction from collected data, and on the validation of such reconstructions [Koene, 2008], as well as fusion of functional and structural data and possible correlational mapping between the two. Comparative investigations may also seek to answer persistent questions about the possible subjective consequences of one procedure of data acquisition and emulation versus another, in terms of self-continuity and other psychological, ethical and philosophical implications.

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